

Thursday, June 28, 2012, 10:47 AM

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*      STAAD.Pro
*      Version 2007   Build 04
*      Proprietary Program of
*      Research Engineers, Intl.
*      Date=   JUN 28, 2012
*      Time=   9:46:55
*
*      USER ID: RNS _ASSOCIATES
*****

1. STAAD SPACE DXF IMPORT OF DRAWING2.DXF
INPUT FILE: 20120627 PIPE SLEEPER 026-028.STD
2. START JOB INFORMATION
3. ENGINEER DATE 27-JUN-12
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. UNIT METER KN
7. JOINT COORDINATES
8. 1 31.1365 0 -27.5421; 2 36.8665 0 -27.5421
9. MEMBER INCIDENCES
10. 1 1 2
11. DEFINE MATERIAL START
12. ISOTROPIC CONCRETE
13. E 2.17185E+007
14. POISSON 0.17
15. DENSITY 23.5616
16. ALPHA 1E-005
17. DAMP 0.05
18. END DEFINE MATERIAL
19. MEMBER PROPERTY AMERICAN
20. 1 PRIS YD 1.64 ZD 0.4
21. CONSTANTS
22. MATERIAL CONCRETE ALL
23. SUPPORTS
24. 1 PINNED
25. 2 FIXED BUT KMZ 0.0001 KMY 0.0001 KMZ 0.0001
26. ***** DEAD LOAD *****
27. LOAD 1 LOADTYPE NONE TITLE DL
28. SELFWEIGHT Y -1
29. MEMBER LOAD
30. 1 UNI GY -120.833
31. ***** FRICTIONAL LOAD *****
32. LOAD 2 LOADTYPE NONE TITLE FL
33. MEMBER LOAD
34. 1 UNI GZ 31.3
35. ***** LOAD COMBINATION - SERVICE *****
36. LOAD COMB 3 1.0DL+1.0FL
37. 1 1.0 2 1.0
38. ***** LOAD COMBINATION - ULTIMATE *****
39. LOAD COMB 4 1.0DL+1.4FL
40. 1 1.0 2 1.4
41. LOAD COMB 5 1.4DL+1.4FL
42. 1 1.4 2 1.4
43. PERFORM ANALYSIS
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PROBLEM STATISTICS

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 2/ 1/ 2

SOLVER USED IS THE IN-CORE ADVANCED SOLVER

TOTAL PRIMARY LOAD CASES = 2, TOTAL DEGREES OF FREEDOM = 6

44. LOAD LIST 3 TO 5
45. PRINT JOINT DISPLACEMENTS ALL

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
1	3	0.0000	0.0000	0.0000	0.0000	-0.0013	-0.0003
	4	0.0000	0.0000	0.0000	0.0000	-0.0018	-0.0003
	5	0.0000	0.0000	0.0000	0.0000	-0.0018	-0.0005
2	3	0.0000	0.0000	0.0000	0.0000	0.0013	0.0003
	4	0.0000	0.0000	0.0000	0.0000	0.0018	0.0003
	5	0.0000	0.0000	0.0000	0.0000	0.0018	0.0005

***** END OF LATEST ANALYSIS RESULT *****

46. PRINT MEMBER FORCES ALL

MEMBER END FORCES STRUCTURE TYPE = SPACE

ALL UNITS ARE -- KN METE (LOCAL)

MEMBER	LOAD	JT	AXIAL	SHEAR-Y	SHEAR-Z	TORSION	MOM-Y	MOM-Z
1	3	1	0.00	390.47	-89.67	0.00	0.00	0.00
		2	0.00	390.47	-89.67	0.00	0.00	0.00
	4	1	0.00	390.47	-125.54	0.00	0.00	0.00
		2	0.00	390.47	-125.54	0.00	0.00	0.00
	5	1	0.00	546.66	-125.54	0.00	0.00	0.00
		2	0.00	546.66	-125.54	0.00	0.00	0.00

***** END OF LATEST ANALYSIS RESULT *****

47. START CONCRETE DESIGN
48. CODE CP65
PROGRAM CODE REVISION V1.0_CP65/1
49. DESIGN BEAM ALL

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B E A M N O. 1 D E S I G N R E S U L T S - F L E X U R E

LEN - 5730. mm FY - 460. FC - 30. SIZE - 400. X 1640. mm

LEVEL	HEIGHT mm	BAR INFO	FROM mm	TO mm	ANCHOR STA END
1	33.	6- 16 MM	0.	5730.	YES YES

B E A M N O. 1 D E S I G N R E S U L T S - S H E A R

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PROVIDE SHEAR LINKS AS FOLLOWS

FROM -	TO	MAX. SHEAR	LOAD	LINKS	NO.	SPACING C/C
END 1	2387 mm	546.7 kN	5	8 mm	12	217 mm
3342	END 2	546.7 kN	5	8 mm	12	217 mm

*****END OF BEAM DESIGN*****

50. END CONCRETE DESIGN

51. FINISH

***** END OF THE STAAD.Pro RUN *****

**** DATE= JUN 28,2012 TIME= 9:46:55 ****

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*           For questions on STAAD.Pro, please contact           *
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*                                                                 *
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